

Lecture 13

Topologist's Sine Curve

- $\{(x, y) \in \mathbb{R}^2 : x > 0 \rightarrow y = \sin(\frac{1}{x}), x = 0 \rightarrow -1 \leq y \leq 1\}$
- Line at $x = 0$ is included in closure of $\sin(\frac{1}{x})$
- Bounded as well so this set is compact
- Set is connected but not path connected
 - Closure of connected is connected
 - No path exists from line at $x = 0$ to point on curve

Spiral

- Similar set up to topologist's sine curve
- $\frac{t-1}{t}(\cos \theta, \sin \theta)$ for $1 \leq t \leq \infty$
- S^1 is closure
- Spiral is path-connected, but no path exists to points on S^1

More on Cantor Sets

- $C \times C \cong C$
- $C \times C$ has 4 properties

Hawaiian Earrings

- Nested circles that are all tangent at origin, centers lie on x -axis
- Think of making Cantor sets out of tori formed by rotating plane
 - Can be formed along the long axis or the short axis

Arcs

- A 1:1 path
- M is path-connected is equivalent to M is arc-connected
 - Prove this!!

Embedding

- $f : M \rightarrow N$
- f embeds M into N if f is one-to-one to its image and continuous, where $f(M)$ has the inherited topology of N
 - Half open interval to circle is not an embedding
- Can also have isometric embeddings, here distances are preserved perfectly

Hilbert Cubes

- H refers to Hilbert space, Q refers to Hilbert cube
- $H = \{(x_n) \text{ sequences in } \mathbb{R} : \sum x^2 \text{ converges}\}$
 - Hilbert space is a vector space
 - Has inner product, defined similarly to $\mathbb{R}^n$
 - Also has metric (more accurately norm I believe)
- $Q = [0, 1]^\infty = \text{all sequences } (x_1, x_2, x_3, \dots) \text{ each component } x_i \in [0, 1]$

Theorem 1: Every compact metric space embeds into the Hilbert cube

- Every compact metric space contains a countable, dense subset
 - Denote this subset as $\{p_1, p_2, p_3, \dots\}$
- $f : M \hookrightarrow Q$
- For some $x \in M$, we can create the sequence $(d(x, p_1), d(x, p_2), d(x, p_3), \dots) \in Q$

- This subset “locates” x in some manner
- Continuity can be shown because points that are close will have similar sequences
- d on compact M , WLOG d is bounded
 - Take $\min(d, 1)$
 - Not sure what this was for

Isometric Embedding

- Think of a Y shape, it cannot be embedded into $\mathbb{R}^2$
- Start with V shape with path metric
 - Can only have isometric embedding as a line in $\mathbb{R}^2$
- Add the third shaft, effectively 3 V shapes, which won't have an embedding in $\mathbb{R}^2$

Higher Dimensions

- Think of a manifold M
- Start by thinking of path metric on torus, is there some isometric embedding to $\mathbb{R}^m$?
- In general $M^d \hookrightarrow \mathbb{R}^{2d+1}$, as $2d$ is the minimum
- Manifolds have no shafts like Y since manifolds look the same locally
- Nash proved yes, bound is $\mathbb{R}^{\frac{d(d+1)(3d+11)}{2}}$
- How smooth is embedding $\mathcal{N} : M \hookrightarrow \mathbb{R}$
- Corrugation
 - An isometric embedding exists for a sphere to a smaller sphere
 - Result by Nash and Kuiper

Theorem 2: Every compact metric space M is separable & A metric space is separable iff the topology of M , $\mathcal{U}$ has countable basis ($\exists$ countably many open sets $B \subset \mathcal{U}$ such that $\forall U \in \mathcal{U}, U = \text{a union of sets in } B$)

$\Rightarrow$

- Dense subset D
- $B = \{M_r(x) : x \in D \wedge r \in \mathbb{Q}^+\}$
- Define B_U for $U \in \mathcal{U}$ to be all points in $U \cap D$
- $\forall p \in V \subset \mathcal{U}$
- $M_{\frac{s}{2}}(p) \subset V$
- Suppose we have some point not included in some element of B_V
 - It will be engulfed by some nearby neighborhood

$\Leftarrow$

- Pick point in each element of basis
- Forms set that is countable